

## Session 7: Nested Loops & Pattern Printing

Topics: Using nested loops for grids or triangle patterns. Example: Star pyramid pattern.

### Tasks

1.

Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 🇪🇺 symbol.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for(i = 1; i <= 5; i++)
```

```
    {
```

```
        for(j = 1; j <= 5; j++)
```

```
        {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
C:\Users\gajja\OneDrive\Desktop > gcc 2.c -o 2.exe
*****
*****
*****
*****
*****
-----
Process exited after 0.1017 seconds with return value 0
Press any key to continue
```

2.

Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for(i = 1; i <= 5; i++)
```

```
    {
```

```
        for(j = 1; j <= i; j++)
```

```
        {
```

```
            printf("%d ", j);
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

```
C:\Users\gajja\OneDrive\Desk  X + v
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

-----
Process exited after 0.101 seconds with return value 0
Press any key to continue . . . |
```

3.

Create a pattern that prints a pyramid of stars (\*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading.<br><br><em><strong>Hint:</strong> Use spaces to align the stars in the center for each row.</em>

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j, space;
```

```
    for(i = 1; i <= 6; i++)
```

```
    {
```

```
        // Print spaces
```

```
        for(space = 1; space <= 6 - i; space++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        // Print stars
```

```
        for(j = 1; j <= 2 * i - 1; j++)
```

```
        {
```

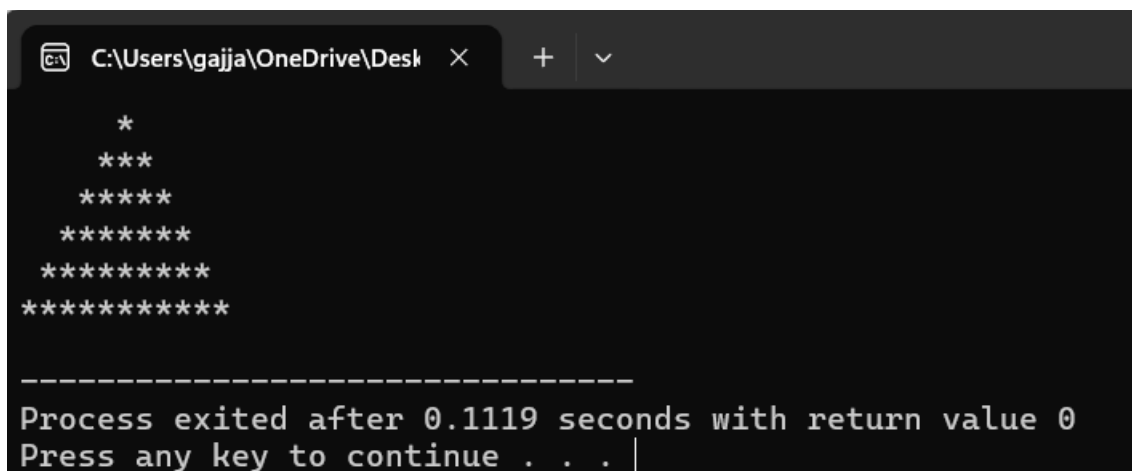
```

        printf("*");
    }

    printf("\n");
}

return 0;
}

```



The screenshot shows a Windows command prompt window with the title bar 'C:\Users\gajja\OneDrive\Desktop'. The window contains the following text:

```

      *
     ***
    *****
   ********
  **********
 **********
 **********

-----
Process exited after 0.1119 seconds with return value 0
Press any key to continue . . .

```

#### 4.

Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).

```
#include <stdio.h>
```

```

int main()
{
    int i, j;

    for(i = 1; i <= 4; i++)
    {
        for(j = 1; j <= 4; j++)

```

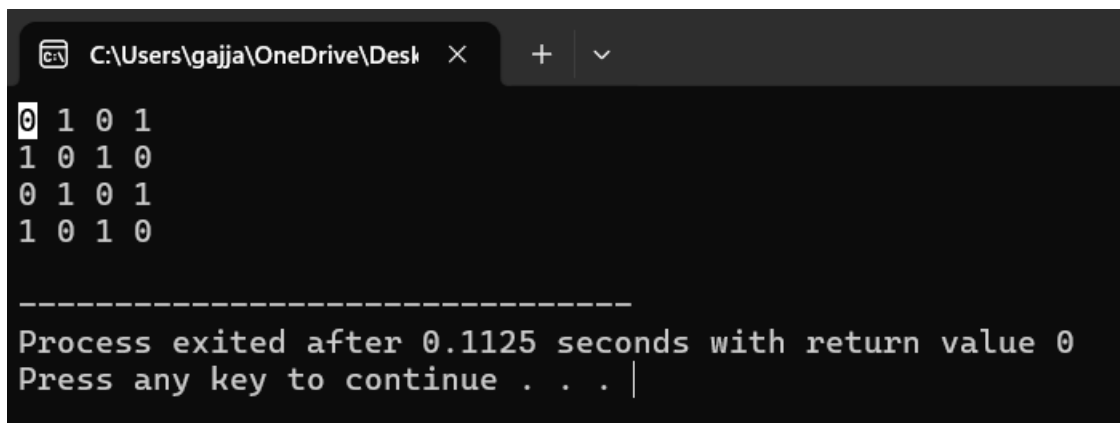
```

{
    if((i + j) % 2 == 0)
    {
        printf("0 ");
    }
    else
    {
        printf("1 ");
    }
}

printf("\n");
}

return 0;
}

```



```

C:\Users\gajja\OneDrive\Desktop >
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0

-----
Process exited after 0.1125 seconds with return value 0
Press any key to continue . . . |

```

5.

Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.

```
#include <stdio.h>
```

```
int main()
{
    int rows, i, j, space;

    printf("Enter number of rows: ");
    scanf("%d", &rows);

    for(i = 1; i <= rows; i++)
    {
        // Print spaces
        for(space = 1; space <= rows - i; space++)
        {
            printf(" ");
        }

        // Print stars
        for(j = 1; j <= 2 * i - 1; j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

C:\Users\gajja\OneDrive\Desk

+

✓

Enter number of rows: 7

```
  *
 ***
*****
*****
*****
*****
*****
```

-----  
Process exited after 3.37 seconds with return value 0  
Press any key to continue . . .